Naga Durga Jagadeesh Lanka

Mechanical Engineer | Mechanical Design & CAD | Robotics, Automation & Fabrication
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PROFESSIONAL SUMMARY

Mechanical engineer completing a Master of Engineering at the University of Cincinnati, with hands on experience in 3D CAD modeling, detailed engineering drawings, and prototype build and test across robotics, automation, and industrial systems. Skilled in SolidWorks, AutoCAD, and Creo, with applied knowledge of GD&T, sheet metal and fabrication drawings, tolerance analysis, FEA, and bill of materials development. Experienced designing and validating automated and robotic systems, from a Fanuc robotic work cell to a LiDAR based autonomous mobile robot. Strong technical documentation and cross functional collaboration skills, comfortable coordinating with manufacturing and procurement to move designs from concept through fabrication and testing.

TECHNICAL SKILLS

CAD & Design: 3D Drawing, 3D Modeling, SolidWorks, AutoCAD, Creo, Onshape, GD&T, Sheet Metal Design, Fabrication & Welding Drawings, Tolerance Analysis, BOM Development, ROBOGUIDE
Analysis & Validation: FEA, Fit & Interference Checks, Root Cause Analysis, Dynamics, DFM and DFA, Prototype Build & Test
Manufacturing & Systems: Robotics & Automation, Hydraulics & Pneumatics, Process Optimization, Quality Control, PLM, Design Reviews
Programming & Tools: Python (rclpy), MATLAB, ROS 2, SQL, Power BI, MS Office (Excel, Word, PowerPoint, Outlook)
Documentation: Technical Reports, Engineering Drawings, BOMs, Design Documentation, Version Control

WORK EXPERIENCE

Mechanical Designer (Remote) | Zouka Solutions, Hyderabad, India *May 2026 – Jul 2026*
- Develop and update 3D CAD models in SolidWorks, AutoCAD, and Creo from design specifications provided by senior engineers through email, Teams, and shared platforms.
- Interpret and convert 2D engineering drawings into accurate 3D models, and generate comprehensive technical drawings with precise dimensions, tolerances, and GD&T annotations.
- Incorporate design changes from virtual design reviews with proper version control, and maintain technical documentation including bills of materials and component lists in Excel and PLM systems.
- Conduct fit checks, interference analysis, and introductory FEA, and coordinate remotely with manufacturing and procurement teams to ensure design feasibility, cost effectiveness, and alignment with project requirements.

Graduate Research Assistant, UPRISE Project (NSF Funded) | University of Cincinnati, OH *Dec 2025 – Apr 2026*
- Reviewed and synthesized 50 plus technical studies for accuracy, relevance, and data quality, producing structured analytical reports for an NSF funded research team while working independently and asynchronously.
- Built a research tracker across 30 plus studies to organize findings and surface gaps, improving team data accessibility and reducing review time.

Design Engineer Intern | GSR Engineering, Hyderabad, India *Jan 2024 – May 2024*
- Prototyped and validated a user driven floor scrubbing machine integrating dry vacuuming and wet mopping, sizing the drivetrain and running fit and interference checks before fabrication.
- Produced fabrication drawings, a bill of materials, and design documentation, and supported process optimization and prototype testing alongside senior engineers.

Engineering Intern | Oil and Natural Gas Corporation Limited (ONGC), India *May 2022 – Jun 2022*
- Analyzed and optimized gas compression capacity control systems, achieving a 15% increase in operational efficiency.
- Conducted research on gas compressor lubrication, reducing maintenance downtime by 20%.
- Evaluated and adjusted gas compressor trip and alarm set points, improving safety protocols and reducing incident rates by 25%.

ENGINEERING PROJECTS

Robotic Work Cell Design and Simulation | Fanuc M10iD/12, ROBOGUIDE *2025*
- Modeled and simulated a complete robotic work cell in SolidWorks and ROBOGUIDE and programmed a Fanuc arm to a repeatability of plus or minus 0.02 mm, validating reach, cycle time, and interference.
- Defined the layout, tooling, and motion sequences for an automated material handling task, documenting the full process for a manufacturing style production scope.

Autonomous Mobile Robot | TurtleBot3, ROS 2, Python, LiDAR, Gazebo *Jan 2025 – Apr 2025*
- Designed and implemented a LiDAR based obstacle avoidance system on a TurtleBot3 mobile platform, validating performance in Gazebo simulation and real world testing.
- Developed control logic in Python with ROS 2, integrating sensor data for reliable autonomous navigation around dynamic obstacles.

Vertical Axis Wind Turbine | CAD, BOM, Fabrication, Field Testing *2024*

- Took a vertical axis wind turbine from CAD models and a bill of materials through a fabricated prototype, then captured field wind measurements to evaluate real performance against design intent.

Advances in Nanotechnology in the Aerospace Industry | University of Cincinnati *Sep 2024 – Dec 2024*

- Researched nanotechnology based materials and self healing nanoparticles for aerospace applications, achieving a 50% weight reduction in wiring, an 8% drag reduction, and a 30% extension in aircraft service life while reducing maintenance costs.

PERSONAL PROJECT

AUREN | AI Desktop Assistant *2026 - Present*

Active development | Python, JavaScript, SQLite, Windows desktop integration

- Developing a voice-driven desktop assistant with a Python controller, interactive web interface, speech recognition, spoken responses and persistent conversation memory in SQLite, using AI-assisted development.
- Integrated OpenAI and Gemini interfaces with provider fallback, persistent cooldowns and redacted diagnostics. Added retrieval of relevant local conversation history and credential isolation through Windows Credential Manager.
- Implemented a Google Search grounding layer with source links and a fallback when live search is unavailable. Local tests cover routing and citation handling. Live grounding validation remains dependent on provider access and quota.
- Packaged a standalone Windows application with a wake listener, restart watchdog and controlled action routing. Continuing work focuses on voice reliability, broader desktop workflows and integration testing.

AI AND SOFTWARE EXPERIENCE

Technical AI Reviewer | Mercor *Jul 2025 - Present*

- Review AI-generated code and technical evaluations against requirements, execution evidence and scoring criteria. Identify inconsistencies between implementation, annotations and test results, and provide actionable revision feedback.

AI Evaluator and Task Author | Handshake AI *2026*

- Developed engineering and scientific AI evaluations involving toleranced drawing interpretation, CAD reconstruction, mechanical systems and advanced STEM reasoning. Prepared reference geometry, worked solutions and quantitative assessment criteria.
- Designed reproducible software and structured-data benchmarks with reference implementations and independent verification. Evaluated root-cause analysis, data-processing logic, checkpoint recovery, replay consistency and dependency handling.
- Assessed model responses for technical accuracy, instruction following and context relevance using execution traces and source evidence. Reconciled ratings, annotations and reviewer feedback to maintain consistent evaluation standards.
- Developed document-based research evaluations by combining technical documents, archival records, maps and tables. Verified answers against source locations and documented the reasoning connecting evidence to conclusions.
- Built playable applications in Defold, Panda3D and Godot, integrating assets, controls and game states. Defined functional acceptance criteria and performed QA of expected behaviour and failure cases.

AI Evaluation Specialist | Outlier *May 2026 – Present*

- Built an AI agent in OpenClaw and configured its execution environment. Developed task specifications, reference trajectories and structured evaluation rubrics for multi-step information retrieval and tool use.
- Analyzed agent execution traces to identify reasoning and tool-use failures. Evaluated privacy boundaries, data minimization and redaction accuracy, and documented evidence supporting assessment decisions.

EDUCATION

Master of Engineering, Mechanical Engineering | University of Cincinnati, OH *Expected Dec 2026*

GPA 3.2 | Coursework: Robotics, Micro & Nano Manufacturing, Quality Control, Robot Control & Design, Industrial Big Data & AI, Intelligence Systems, Engineering Project Management, Effectiveness in Technical Organizations.

Bachelor of Technology, Mechanical Engineering | NIT Manipur, India *2024*

- Thesis (sole author): User-Driven Floor Scrubbing Machine. Designed a chain and bevel gear drivetrain, ran free body and mechanical advantage analysis, and completed a manufacturing and operating cost study against electric units.

CERTIFICATIONS

Certified Associate in Project Management (CAPM): Project Management Institute (PMI)
GRE: 326